

EXPLANATORY NOTES

COMPARABLE SPECIFICATIONS AND CHEMICAL ANALYSIS

Specifications

The following specifications are normally readily available, but any offer will depend upon acceptance of full specification details and/or any other specifications not listed below.

Structural Sections

Quality	Grade	Tensile strength			Min. yield stress		
		lb/in ²	N/mm ²	kg/mm ²	lb/in ²	N/mm ²	kg/mm ²
BS4360	43A	62300/84000	430/580	44/59	38500	265	27
	50A	72500/93000	490/640	50/65	50000	345	36
ASTM	A36	58000/80000	400/550	41/56	36000	250	25
ASTM A572	50	65000 min	450 min	46 min	50000	345	36
Din 17100	RSt. 37.2	52000/74000	360/510	36/52	32500	225	23
	St 44.2	62300/83500	430/580	44/59	38500	265	27
	St 52.3	74000/98500	510/680	52/69	50000	345	36
JIS G 3101	SS 41	58000/74000	402/510	41/52	34000	235	24
	SS 50	72500/88500	490/608	50/62	40000	275	28
JIS G 3106	SM 41	58000/74000	402/510	41/52	34000	235	24
	SM 50	72500/88500	490/608	50/62	45000	314	32

Notes: Sections over 5/8 inch (16mm) up to and including 1 inch (25mm) thick
All figures are approximate and for guidance only
Other specifications are obtainable subject to acceptance of specification (eg. UNI 7070-72 and NFA 35-501)

Chemical analyses

Specification	Grade	Composition (Maxima unless stated)								
Quality		C	Si	Mn	P	S	Nb	V	N1	Others
BS4360 (1986)	43A	0.25	0.50	1.60	0.05	0.05	--	--	--	--
	50B	0.20	0.50	1.50	0.05	0.05	0.003/0.10	0.003/0.10	--	--
ASTM A36 (1984)		0.26	--	--	0.04	0.05	--	--	--	Cu 0.20 (min)
ASTM A572 (1984)	50	0.23	0.40	1.35	0.04	0.05	--	--	--	Cu 0.20 (min)
JIS 3101 (1976)	SS 41	--	--	--	0.05	0.05	--	--	--	--
	SS 50	--	--	--	0.05	0.05	--	--	--	--
JIS 3106 (1976)	SM 41 A	0.25	--	2.5 x C min	0.04	0.04	--	--	--	--
	SM 41 B	0.22	0.35	0.6 - 1.20	0.04	0.04	--	--	--	--
	SM 41 C	0.18	0.35	1.40	0.04	0.04	--	--	--	--
	SM 50 A	0.22	0.55	1.50	0.04	0.04	--	--	--	--
	SM 50 B	0.20	0.55	1.50	0.04	0.04	--	--	--	--
	SM 50 C	0.18	0.55	1.50	0.04	0.04	--	--	--	--